

Asymptotic Normality for Tamed Stochastic Mirror Descent in Risk-Budgeting Portfolios: A Rigorous Partial Solution

Status of the result. This paper gives a *rigorous partial solution* to the central-limit-theorem problem posed by Arnaiz Iglesias, Cetingoz, and Frikha. It proves raw-iterate and arithmetic-average central limit theorems for $\gamma_n = \gamma_0 n^{-\alpha}$, $1/2 < \alpha < 1$, when the optimizer lies strictly inside the artificial ℓ_1 ball and the population objective is locally nondegenerate and sufficiently smooth. It does not treat the active-boundary case $m = \|y^*\|_1$, the critical $1/n$ step size, or every nonsmooth risk measure covered by the source paper.

Abstract

Arnaiz Iglesias, Cetingoz, and Frikha proposed a tamed stochastic mirror-descent algorithm for risk-budgeting portfolios and identified a central limit theorem for its asymptotic error as an open direction. We resolve the regular strict-interior regime under explicit local regularity assumptions. If $m > \|y^*\|_1$, the radial part of the KL proximal map is almost surely inactive after a finite random time. In the coordinates $w = (\xi, \log y)$, the remaining update is then exactly a Euclidean Robbins–Monro recursion. A cutoff-and-restart localization argument permits stochastic-approximation moment estimates without any global moment assumption on the original iterates. For $1/2 < \alpha < 1$, we obtain $\gamma_n^{-1/2}(z_n - z^*) \Rightarrow \mathcal{N}(0, \Sigma_{\text{raw}})$, where Σ_{raw} solves an explicit Lyapunov equation. For the ordinary arithmetic average, we obtain the Polyak–Ruppert limit $\sqrt{n}(\hat{z}_n - z^*) \Rightarrow \mathcal{N}(0, B^{-1}CB^{-1})$. The KL geometry and the taming factor affect the raw covariance but cancel from the averaged covariance. We also prove that the source paper’s step-size-weighted average has a convergent random-series numerator. No model-specific Gaussian or non-Gaussian assertion for that limit is made.

1 Relation to the source paper and notation

The source paper [1] studies the stochastic mirror-descent recursion below, proves almost-sure convergence under its global hypotheses, and derives a nonasymptotic bound for a step-size-weighted average. Its conclusion explicitly lists a central limit theorem for the stochastic algorithm as a future direction. The result here solves that direction in the regular strict-interior regime and is partial overall.

Three proof points are essential. Almost-sure convergence cannot by itself be upgraded to an unconditional L^2 rate, so Appendix A uses a cutoff recursion started at a deterministic time and transfers its estimates on events whose probabilities tend to one. The quadratic Taylor remainder under arithmetic averaging is controlled through this localized moment estimate, not by summing unrelated $O_{\mathbb{P}}$ statements. Finally, for the source paper’s step-size-weighted average we prove only the random-series convergence that follows from the recursion.

Let X be an $\mathbb{R}^d$ -valued return vector and let $b = (b_1, \dots, b_d) \in (0, \infty)^d$ with $\sum_i b_i = 1$. Suppose the relevant risk functional admits the stochastic representation

$$g(r_\rho(y)) = \min_{\xi \in \mathbb{R}} \mathbb{E} L(\xi, -\langle y, X \rangle).$$

For $z = (\xi, y) \in \mathbb{R} \times (0, \infty)^d$, define

$$H(z, x) = L(\xi, -\langle y, x \rangle) - \sum_{i=1}^d b_i \log y_i,$$

$$h(z) = \mathbb{E}H(z, X).$$

Let $z^* = (\xi^*, y^*)$ be the unique minimizer of h . We write

$$\kappa(y) = \left(\min_{1 \leq i \leq d} y_i \right) \wedge 1.$$

For a radius $m \geq \|y^*\|_1$, the stochastic mirror-descent recursion of [1] is

$$\xi_{n+1} = \xi_n - \gamma_{n+1} \partial_\xi H(z_n, X_{n+1}), \quad (1)$$

$$y_{n+1} = P_{y_n}^m(\gamma_{n+1} \kappa(y_n) \nabla_y H(z_n, X_{n+1})), \quad (2)$$

where, componentwise,

$$P_y^m(v) = \begin{cases} y \odot e^{-v}, & \|y \odot e^{-v}\|_1 \leq m, \\ m \frac{y \odot e^{-v}}{\|y \odot e^{-v}\|_1}, & \|y \odot e^{-v}\|_1 > m. \end{cases} \quad (3)$$

The goal is to identify a normalization and limiting covariance for the error of this recursion.

Choose a measurable stochastic gradient (or a measurable subgradient selection when L is nonsmooth)

$$\mathcal{G}(z, x) := \begin{pmatrix} L_1(\xi, -y^\top x) \\ -x L_2(\xi, -y^\top x) - b \oslash y \end{pmatrix}, \quad (4)$$

where $b \oslash y = (b_i/y_i)_{i=1}^d$. We assume throughout that this selection is locally unbiased: on a neighborhood of z^* ,

$$\mathbb{E}[\mathcal{G}(z, X)] = \nabla h(z), \quad \mathbb{E}[\mathcal{G}(z^*, X)] = 0. \quad (5)$$

Set

$$B := \nabla^2 h(z^*), \quad C := \mathbb{E}[\mathcal{G}(z^*, X) \mathcal{G}(z^*, X)^\top]. \quad (6)$$

The matrix C is the covariance of the stochastic gradient at the optimum, because its mean is zero.

Let

$$D_\star := \text{diag}(y_1^*, \dots, y_d^*), \quad \kappa_\star := \kappa(y^*), \quad (7)$$

and define the three positive diagonal matrices

$$R_\star := \text{diag}(1, \kappa_\star I_d), \quad T_\star := \text{diag}(1, D_\star), \quad Q_\star := T_\star R_\star = \text{diag}(1, \kappa_\star D_\star). \quad (8)$$

The matrix $T_\star$ is the Jacobian of $(\xi, \theta) \mapsto (\xi, e^\theta)$ at $(\xi^*, \log y^*)$, while $R_\star$ contains the taming factor. Their product $Q_\star$ is the local primal mirror preconditioner.

2 Assumptions

The following are local assumptions. In particular, they do not require the sample loss L to be globally smooth.

Assumption 2.1 (Convergence and strict radial slack). The recursion is well defined and $z_n \rightarrow z^*$ almost surely. Moreover,

$$m > \|y^*\|_1.$$

Assumption 2.2 (Local curvature). There is a neighborhood U of z^* on which h is twice differentiable, $B = \nabla^2 h(z^*)$ is positive definite, and, for some $K < \infty$,

$$\|\nabla h(z) - B(z - z^*)\| \leq K\|z - z^*\|^2, \quad z \in U. \quad (9)$$

Assumption 2.3 (Local moments and continuity of the oracle). For some $\delta > 0$ and some random variable $M(X)$,

$$\mathbb{E}[M(X)^{2+\delta}] < \infty, \quad \sup_{z \in U} \|\mathcal{G}(z, X)\| \leq M(X) \quad \text{a.s.} \quad (10)$$

After shrinking U if necessary,

$$\mathbb{E}\|\mathcal{G}(z, X) - \mathcal{G}(z^*, X)\|^2 \rightarrow 0 \quad \text{as } z \rightarrow z^*. \quad (11)$$

The samples $X_1, X_2, \dots$ are i.i.d. and independent of z_0 .

Assumption 2.4 (Step sizes). For constants $\gamma_0 > 0$ and $\alpha \in (1/2, 1)$,

$$\gamma_n = \gamma_0 n^{-\alpha}. \quad (12)$$

Assumption 2.1 is supplied by the almost-sure convergence theorem in the source paper under its global hypotheses. Assumptions 2.2 and 2.3 are the usual nondegeneracy, moment, and local regularity conditions for a stochastic-approximation CLT. The quadratic remainder in (9) follows, for example, from a locally Lipschitz Hessian. Nonsmooth sample losses such as Expected Shortfall remain admissible provided averaging over X makes h twice differentiable and (11) holds.

3 Main central limit theorems

Define

$$A := Q_* B, \quad S := Q_* C Q_*. \quad (13)$$

Although A need not be symmetric, it is positive stable. Indeed,

$$Q_*^{-1/2} A Q_*^{1/2} = Q_*^{1/2} B Q_*^{1/2} \succ 0. \quad (14)$$

Consequently, the continuous Lyapunov equation below has a unique symmetric positive semidefinite solution.

Theorem 3.1 (Raw-iterate CLT). *Under Assumptions 2.1–2.4,*

$$\gamma_n^{-1/2}(z_n - z^*) \implies \mathcal{N}(0, \Sigma_{\text{raw}}), \quad (15)$$

where Σ_{raw} is the unique solution of

$$A \Sigma_{\text{raw}} + \Sigma_{\text{raw}} A^\top = S. \quad (16)$$

Equivalently,

$$\Sigma_{\text{raw}} = \int_0^\infty e^{-At} S e^{-A^\top t} dt. \quad (17)$$

Since $\gamma_n = \gamma_0 n^{-\alpha}$, this may also be written as

$$n^{\alpha/2}(z_n - z^\star) \implies \mathcal{N}(0, \gamma_0 \Sigma_{\text{raw}}). \quad (18)$$

Thus the requested normalization is $a_n = \gamma_n^{-1/2}$, and the mirror map and taming enter the covariance precisely through $\mathbf{Q}_\star = \text{diag}(1, \kappa_\star D_\star)$.

The next result uses the ordinary arithmetic average, not the step-size-weighted average used for the paper's nonasymptotic objective bound:

$$\hat{z}_n := \frac{1}{n} \sum_{k=0}^{n-1} z_k. \quad (19)$$

Theorem 3.2 (Polyak–Ruppert CLT). *Under Assumptions 2.1–2.4,*

$$\sqrt{n}(\hat{z}_n - z^\star) \implies \mathcal{N}(0, \Sigma_{\text{PR}}), \quad (20)$$

where

$$\boxed{\Sigma_{\text{PR}} = B^{-1} C B^{-1}}. \quad (21)$$

In particular, the first-order covariance of the averaged estimator is independent of the KL mirror geometry, the value of $\kappa_\star$, and the constant γ_0 .

4 Proof of the main theorems

4.1 The radial KL projection is eventually inactive

Lemma 4.1 (Eventual inactivity). *Under Assumptions 2.1, 2.3, and 2.4, with probability one there is a finite random index N such that, for all $n \geq N$,*

$$y_{n+1} = y_n \odot \exp(-\gamma_{n+1} \kappa(y_n) \nabla_y H(z_n, X_{n+1})), \quad (22)$$

where the exponential is componentwise. In other words, the second branch in (3) is used only finitely many times.

Proof. Let $r = m - \|y^\star\|_1 > 0$. Since $y_n \rightarrow y^\star$ almost surely, eventually

$$\|y_n\|_1 \leq \|y^\star\|_1 + \frac{r}{3} = m - \frac{2r}{3}. \quad (23)$$

On the same event, $z_n \in U$ eventually. By Markov's inequality and (10), for every $\varepsilon > 0$,

$$\sum_{n=1}^{\infty} \mathbb{P}(\gamma_{n+1} M(X_{n+1}) > \varepsilon) \leq \varepsilon^{-(2+\delta)} \mathbb{E}[M^{2+\delta}] \sum_{n=1}^{\infty} \gamma_{n+1}^{2+\delta} < \infty.$$

The Borel–Cantelli lemma gives $\gamma_{n+1} M(X_{n+1}) \rightarrow 0$ almost surely. Put

$$v_{n+1} = \gamma_{n+1} \kappa(y_n) \nabla_y H(z_n, X_{n+1}).$$

Then $\|v_{n+1}\|_\infty \leq \gamma_{n+1} M(X_{n+1})$ eventually, because $0 < \kappa \leq 1$. Hence

$$\|y_n \odot e^{-v_{n+1}}\|_1 \leq e^{\|v_{n+1}\|_\infty} \|y_n\|_1 < m$$

for all sufficiently large n , by (23). The first branch of (3) then applies. $\square$

Remark 4.2. *The strict inequality $m > \|y^\star\|_1$ is exactly what makes the radial constraint asymptotically invisible. If $m = \|y^\star\|_1$, stochastic radial rescaling can remain active near the limit; that is a projected stochastic-approximation problem with a different, potentially non-Gaussian boundary limit.*

4.2 Exact reduction to Robbins–Monro in logarithmic coordinates

Set

$$w = (\xi, \theta), \quad \theta_i = \log y_i, \quad \psi(w) = (\xi, e^\theta). \quad (24)$$

Let $w^\star = (\xi^\star, \log y^\star)$. Define

$$R(y) := \text{diag}(1, \kappa(y)I_d), \quad \Phi(w, x) := R(e^\theta)\mathcal{G}(\psi(w), x). \quad (25)$$

By Lemma 4.1, after an almost surely finite number of iterations, (1)–(2) are exactly

$$w_{n+1} = w_n - \gamma_{n+1}\Phi(w_n, X_{n+1}). \quad (26)$$

No Taylor expansion of the mirror step is involved: taking logarithms converts multiplication by the exponential into addition.

The mean field is

$$\phi(w) := \mathbb{E}\Phi(w, X) = R(e^\theta)\nabla h(\psi(w)). \quad (27)$$

The map κ need not be differentiable at ties among the coordinates of $y^\star$, or at $\min_i y_i^\star = 1$. This causes no difficulty. Since $\nabla h(z^\star) = 0$, the variation of $R(y)$ is multiplied by a vanishing factor. The Lipschitz continuity of κ and Assumption 2.2 imply

$$\phi(w) = J(w - w^\star) + O(\|w - w^\star\|^2), \quad J = R_\star B T_\star. \quad (28)$$

Moreover, the noise covariance at equilibrium is

$$\Omega := \mathbb{E}[\Phi(w^\star, X)\Phi(w^\star, X)^\top] = R_\star C R_\star. \quad (29)$$

The matrix J is positive stable. Indeed,

$$T_\star J T_\star^{-1} = T_\star R_\star B = Q_\star B = A, \quad (30)$$

and A is similar to the positive definite matrix in (14).

The following localized version of the classical stochastic-approximation central limit theorems [2, 3, 5] is proved in Appendix A. The cutoff-and-restart proof is the main technical repair: almost-sure convergence is never used to infer an unconditional second-moment estimate for the original process.

Theorem 4.3 (Localized Robbins–Monro and averaging CLTs). *Let $(\mathcal{F}_n)_{n \geq 0}$ be a filtration, let W_n be $\mathcal{F}_n$ -adapted, and let ζ_{n+1} be independent of $\mathcal{F}_n$, with a law not depending on n . Assume $W_n \rightarrow w_0$ almost surely and that, after an almost surely finite random index, the recursion has the form*

$$W_{n+1} = W_n - \gamma_{n+1}\Phi(W_n, \zeta_{n+1}), \quad \gamma_n = \gamma_0 n^{-\alpha}, \quad \frac{1}{2} < \alpha < 1.$$

Assume, in a neighborhood of w_0 , that

- (i) $\phi(w) := \mathbb{E}\Phi(w, \zeta)$ satisfies $\phi(w) = J(w - w_0) + O(\|w - w_0\|^2)$ with J positive stable;

(ii) for some $\delta > 0$, a neighborhood V_0 of w_0 , and a random variable M with $\mathbb{E}[M^{2+\delta}] < \infty$,
 $\sup_{w \in V_0} \|\Phi(w, \zeta)\| \leq M$ almost surely;

(iii) $\mathbb{E} \|\Phi(w, \zeta) - \Phi(w_0, \zeta)\|^2 \rightarrow 0$ as $w \rightarrow w_0$.

Let $\Omega = \mathbb{E}[\Phi(w_0, \zeta)\Phi(w_0, \zeta)^\top]$. Then

$$\gamma_n^{-1/2}(W_n - w_0) \Rightarrow \mathcal{N}(0, V), \quad JV + VJ^\top = \Omega, \quad (31)$$

$$\sqrt{n} \left(\frac{1}{n} \sum_{k=0}^{n-1} W_k - w_0 \right) \Rightarrow \mathcal{N}(0, J^{-1}\Omega J^{-\top}). \quad (32)$$

The Lyapunov equation has a unique symmetric positive-semidefinite solution. In addition,

$$\sum_{n=1}^{\infty} \gamma_{n+1} \|W_n - w_0\|^2 < \infty \quad \text{almost surely.} \quad (33)$$

Because ψ is smooth and R is locally bounded and continuous, Assumption 2.3 transfers to Φ : after shrinking the log-coordinate neighborhood, Φ has a $(2 + \delta)$ envelope and is L^2 -continuous at w^* . Thus Theorem 4.3 applies to (26) and gives

$$\gamma_n^{-1/2}(w_n - w^*) \Rightarrow \mathcal{N}(0, V_w), \quad JV_w + V_w J^\top = \Omega. \quad (34)$$

The raw CLT makes $\gamma_n^{-1/2}(w_n - w^*)$ tight. Hence the quadratic remainder in

$$z_n - z^* = T_\star(w_n - w^*) + O(\|w_n - w^*\|^2) \quad (35)$$

is $o_{\mathbb{P}}(\sqrt{\gamma_n})$. Thus $\Sigma_{\text{raw}} = T_\star V_w T_\star$. Multiplying the Lyapunov equation in (34) on the left and right by $T_\star$ and using (30) gives

$$\begin{aligned} A\Sigma_{\text{raw}} + \Sigma_{\text{raw}}A^\top &= T_\star \Omega T_\star \\ &= T_\star R_\star C R_\star T_\star = Q_\star C Q_\star = S. \end{aligned}$$

This proves Theorem 3.1.

For averaging, Theorem 4.3 gives

$$\sqrt{n} \left(\frac{1}{n} \sum_{k=0}^{n-1} w_k - w^* \right) \Rightarrow \mathcal{N}(0, J^{-1}\Omega J^{-\top}). \quad (36)$$

Appendix A also proves, by the same localization,

$$\frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} \|w_k - w^*\|^2 \rightarrow 0 \quad \text{in probability.} \quad (37)$$

Averaging (35) therefore permits the linear delta method. Finally,

$$\begin{aligned} T_\star J^{-1}\Omega J^{-\top} T_\star &= B^{-1}R_\star^{-1}(R_\star C R_\star)R_\star^{-1}B^{-1} \\ &= B^{-1}CB^{-1}, \end{aligned} \quad (38)$$

which proves Theorem 3.2.

5 The source paper's step-size-weighted average

The source paper studies

$$\bar{z}_n^\gamma = \frac{\sum_{k=1}^n \gamma_k z_{k-1}}{T_n}, \quad T_n = \sum_{k=1}^n \gamma_k. \quad (39)$$

This is not the Polyak–Ruppert average.

Proposition 5.1 (Convergent numerator). *Under Assumptions 2.1–2.4, there is a finite random vector W_γ such that*

$$T_n(\bar{z}_n^\gamma - z^*) = \sum_{k=1}^n \gamma_k(z_{k-1} - z^*) \longrightarrow W_\gamma \quad \text{almost surely.} \quad (40)$$

Moreover, $T_n \sim \gamma_0 n^{1-\alpha}/(1-\alpha)$. No Gaussianity assertion for W_γ follows from the stated local assumptions.

Proof. After the finite time in Lemma 4.1, write $e_k = w_k - w^*$ and

$$e_k = e_{k-1} - \gamma_k \{J e_{k-1} + r(e_{k-1}) + \eta_k\}, \quad (41)$$

where $r(e) = O(\|e\|^2)$ and $\eta_k = \Phi(w_{k-1}, X_k) - \phi(w_{k-1})$ is a martingale difference. The summability conclusion (33) gives $\sum_k \gamma_k \|e_{k-1}\|^2 < \infty$ almost surely. Hence $\sum_k \gamma_k r(e_{k-1})$ converges absolutely.

To treat the martingale series without a global moment condition, use the cutoff processes in Appendix A. For each fixed cutoff process, the global second-moment bound and $\sum_k \gamma_k^2 < \infty$ imply that $\sum_k \gamma_k \eta_k$ converges almost surely, indeed in L^2 . On the event A_N in the transfer argument, the cutoff and original series agree from time N onward. Since $\mathbb{P}(\bigcup_N A_N) = 1$, the original martingale series converges almost surely.

Summing (41) now shows that $\sum_k \gamma_k e_{k-1}$ converges, because J is invertible. Finally,

$$\psi(w^* + e) - z^* = T_* e + O(\|e\|^2),$$

so $\sum_k \gamma_k(z_{k-1} - z^*)$ also converges. This is (40). $\square$

Remark 5.2 (Why a universal Gaussian claim would be false). *The following example concerns the broader Robbins–Monro class; it is not claimed to be a risk-budgeting instance. Let*

$$e_k = e_{k-1} - \gamma_k(ae_{k-1} + \varepsilon_k), \quad a > 0,$$

where $e_0 = 0$ and the ε_k are i.i.d., bounded, centered, and satisfy $\mathbb{E}\varepsilon_1^3 \neq 0$. Standard stochastic-approximation estimates give $e_n \rightarrow 0$ almost surely, and

$$aT_n \bar{e}_n^\gamma = -e_n - \sum_{k=1}^n \gamma_k \varepsilon_k \longrightarrow -\sum_{k=1}^{\infty} \gamma_k \varepsilon_k.$$

The series converges in L^3 , and its third centered moment is

$$-\mathbb{E}[\varepsilon_1^3] \sum_{k=1}^{\infty} \gamma_k^3 \neq 0.$$

Thus its limit is non-Gaussian. This shows only that the local stochastic-approximation assumptions do not force a Gaussian limit for step-size weighting; it does not establish non-Gaussianity for the risk-budgeting model.

6 Covariance in model primitives

When L is twice differentiable and differentiation may be interchanged with expectation, the matrices in the preceding theorems can be written explicitly. Let

$$T_X^* := -y^{*\top} X$$

and evaluate every derivative of L below at (ξ^*, T_X^*) . Then

$$B = \begin{pmatrix} \mathbb{E}L_{11} & -\mathbb{E}[L_{12}X^\top] \\ -\mathbb{E}[L_{21}X] & \mathbb{E}[L_{22}XX^\top] + \text{diag}\left(\frac{b_i}{(y_i^*)^2}\right)_{i=1}^d \end{pmatrix}. \quad (42)$$

Furthermore,

$$C = \mathbb{E} \left[\begin{pmatrix} L_1 \\ -XL_2 - b \odot y^* \end{pmatrix} \begin{pmatrix} L_1 \\ -XL_2 - b \odot y^* \end{pmatrix}^\top \right]. \quad (43)$$

Equations (8), (13), and (16) then give the complete raw-iterate covariance in terms of (L, X, b, z^*) and the local taming factor. The primitives (g, ρ) enter through their stochastic representation L and through the optimizer z^* .

For nonsmooth L , formulas (6) remain the correct definitions: B is the Hessian of the *population* objective and C is the covariance of the selected stochastic oracle at the optimum. This is particularly useful for quantile- and Expected-Shortfall-type losses, where L is piecewise linear but the expectation is smooth under a density condition.

6.1 A concrete model: volatility

For volatility the source paper takes

$$L(\xi, x) = (x - \xi)^2.$$

Let $\mu = \mathbb{E}X$ and $M_2 = \mathbb{E}[XX^\top]$. Then

$$B = 2 \begin{pmatrix} 1 & \mu^\top \\ \mu & M_2 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & \text{diag}(b_i/(y_i^*)^2) \end{pmatrix}. \quad (44)$$

This matrix is automatically positive definite, since for $(a, v) \neq 0$,

$$(a, v)^\top B(a, v) = 2\mathbb{E}(a + v^\top X)^2 + \sum_{i=1}^d \frac{b_i}{(y_i^*)^2} v_i^2 > 0.$$

If $\mathbb{E}\|X\|^{4+2\delta} < \infty$ for some $\delta > 0$, the local moment and L^2 -continuity assumptions hold. Consequently, whenever the almost-sure convergence premise is available and $m > \|y^*\|_1$, the CLTs above apply to the volatility model. At the optimum, $\xi^* = -\mu^\top y^*$, and C is the covariance of

$$\begin{pmatrix} 2(\xi^* + y^{*\top} X) \\ 2X(\xi^* + y^{*\top} X) - b \odot y^* \end{pmatrix}.$$

7 Inference for the normalized risk-budgeting portfolio

The reported portfolio is

$$u^\star = \frac{y^\star}{\mathbf{1}^\top y^\star}. \quad (45)$$

Let $s_\star = \mathbf{1}^\top y^\star$ and

$$N_\star := \frac{1}{s_\star} (I_d - u^\star \mathbf{1}^\top), \quad (46)$$

which is the Jacobian of $y \mapsto y/(\mathbf{1}^\top y)$ at $y^\star$. For any $(d+1) \times (d+1)$ covariance matrix Σ , denote its y - y block by $[\Sigma]_{yy}$.

Corollary 7.1 (Portfolio-weight CLTs). *Let $u_n = y_n/(\mathbf{1}^\top y_n)$ and $\hat{u}_n = \hat{y}_n/(\mathbf{1}^\top \hat{y}_n)$. Under the assumptions of Theorems 3.1 and 3.2,*

$$\gamma_n^{-1/2}(u_n - u^\star) \implies \mathcal{N}\left(0, N_\star[\Sigma_{\text{raw}}]_{yy} N_\star^\top\right), \quad (47)$$

$$\sqrt{n}(\hat{u}_n - u^\star) \implies \mathcal{N}\left(0, N_\star[B^{-1}CB^{-1}]_{yy} N_\star^\top\right). \quad (48)$$

Both covariance matrices are singular in the direction $\mathbf{1}$, as they must be because portfolio weights sum to one.

8 Scope and unresolved cases

Theorems 3.1 and 3.2 completely resolve the regime specified by Assumptions 2.1–2.4. They constitute a rigorous partial solution to the broader problem in the source paper. Four substantive extensions remain outside the present result.

- (1) **Active radial boundary.** If $m = \|y^\star\|_1$, radial projection need not become inactive. A projected stochastic-approximation analysis is then required, and the limiting law may differ from the interior Lyapunov formula.
- (2) **Degenerate curvature.** Strict convexity of h does not imply $\nabla^2 h(z^\star) \succ 0$. Degenerate minima can have nonstandard rates and non-Gaussian limits.
- (3) **Model-by-model verification.** For Expected Shortfall and other piecewise-linear losses, L^2 continuity of the oracle follows under suitable no-atom and moment conditions, but the quadratic population-drift expansion must still be checked from the return distribution.
- (4) **The critical $1/n$ step size.** The case $\alpha = 1$ has the usual Fabian drift correction and requires a spectral lower bound depending on γ_0 ; it is not included here.

A A rigorous localized stochastic-approximation theorem

This appendix proves Theorem 4.3. The purpose of the cutoff-and-restart argument is to avoid the invalid implication “almost-sure convergence $\Rightarrow$ an unconditional L^2 rate.”

A.1 Deterministic product estimates

Let J be positive stable and

$$\Pi_{n,k} = \prod_{j=k+1}^n (I - \gamma_j J), \quad 0 \leq k \leq n,$$

with the empty product equal to I . Write $t_n = \sum_{j=1}^n \gamma_j$.

Lemma A.1 (Fundamental-matrix bounds). *For $\gamma_n = \gamma_0 n^{-\alpha}$ with $0 < \alpha < 1$, there are constants $c, C > 0$ such that, for all sufficiently large $n \geq k$,*

$$\|\Pi_{n,k}\| \leq C e^{-c(t_n - t_k)}, \quad (49)$$

$$\sum_{k=1}^n \gamma_k^2 e^{-c(t_n - t_k)} \leq C \gamma_n, \quad (50)$$

$$\max_{1 \leq k \leq n} \frac{\gamma_k}{\sqrt{\gamma_n}} e^{-c(t_n - t_k)} \leq C \sqrt{\gamma_n}. \quad (51)$$

Proof. Let $P \succ 0$ solve $J^\top P + PJ = I$. For all sufficiently small γ ,

$$(I - \gamma J)^\top P (I - \gamma J) \preceq (1 - c_0 \gamma) P$$

for some $c_0 > 0$. Iteration yields (49).

For (50), split the sum at $n/2$. If $k > n/2$, then $\gamma_k \leq 2^\alpha \gamma_n$ and $t_n - t_k \geq (n - k)\gamma_n$, so

$$\sum_{k > n/2} \gamma_k^2 e^{-c(t_n - t_k)} \leq C \gamma_n^2 \sum_{r=0}^{\infty} e^{-cr\gamma_n} \leq C \gamma_n.$$

If $k \leq n/2$, then $t_n - t_k \geq c_1 n^{1-\alpha}$, and the resulting exponential factor dominates every polynomial. The proof of (51) is identical: on $k > n/2$ the expression is at most $C \sqrt{\gamma_n}$, while the earlier terms are exponentially small. $\square$

Lemma A.2 (Discrete covariance limit). *Let Ω be symmetric positive semidefinite and define*

$$V_n = \frac{1}{\gamma_n} \sum_{k=1}^n \gamma_k^2 \Pi_{n,k} \Omega \Pi_{n,k}^\top. \quad (52)$$

Then $V_n \rightarrow V$, where V is the unique solution of

$$JV + VJ^\top = \Omega. \quad (53)$$

Proof. Lemma A.1 shows that (V_n) is bounded. Its exact recursion is

$$V_{n+1} = \frac{\gamma_n}{\gamma_{n+1}} (I - \gamma_{n+1} J) V_n (I - \gamma_{n+1} J)^\top + \gamma_{n+1} \Omega. \quad (54)$$

Because $\alpha < 1$, $\gamma_n / \gamma_{n+1} = 1 + o(\gamma_{n+1})$. Expanding and using boundedness,

$$V_{n+1} = V_n + \gamma_{n+1} \{\Omega - JV_n - V_n J^\top + o(1)\}.$$

Let $\mathcal{L}(D) = JD + DJ^\top$. After vectorization, $\mathcal{L}$ is represented by $I \otimes J + J \otimes I$; its eigenvalues are sums of eigenvalues of J and therefore have positive real parts. Let V solve $\mathcal{L}(V) = \Omega$ and set $D_n = V_n - V$. Then

$$D_{n+1} = (I - \gamma_{n+1}\mathcal{L})D_n + \gamma_{n+1}\epsilon_n, \quad \epsilon_n \rightarrow 0.$$

The Lyapunov argument of Lemma A.1, now on the finite-dimensional matrix space, bounds products of $I - \gamma_j\mathcal{L}$ by $Ce^{-c(t_n - t_k)}$. Variation of constants makes the initial term vanish. In the forcing sum, finitely many early indices vanish exponentially, while the tail is bounded by $C \sup_{k > K} \|\epsilon_k\|$ because $\sum_{k=K+1}^n \gamma_k e^{-c(t_n - t_k)} \leq C$. Letting first $n \rightarrow \infty$ and then $K \rightarrow \infty$ proves $D_n \rightarrow 0$. Equivalently,

$$V = \int_0^\infty e^{-Jt} \Omega e^{-J^\top t} dt.$$

□

A.2 A globally stable cutoff recursion

Translate so that $w_0 = 0$ and write locally

$$\phi(e) = Je + r(e), \quad \|r(e)\| \leq K \|e\|^2. \quad (55)$$

Choose $\rho > 0$ small enough that the local assumptions hold on $\overline{B}_{2\rho}$. Let $\chi : \mathbb{R}^q \rightarrow [0, 1]$ be continuous, equal to one on $\overline{B}_\rho$, and equal to zero outside $B_{2\rho}$. Define

$$\tilde{\Phi}(e, x) = \begin{cases} Je + \chi(e)(\Phi(e, x) - Je), & \|e\| < 2\rho, \\ Je, & \|e\| \geq 2\rho. \end{cases} \quad (56)$$

Its mean field is

$$\tilde{\phi}(e) = Je + \tilde{r}(e), \quad \tilde{r}(e) = \chi(e)r(e), \quad \|\tilde{r}(e)\| \leq K \|e\|^2. \quad (57)$$

Lemma A.3 (Moment rate and almost-sure summability). *Consider*

$$e_{n+1} = e_n - \gamma_{n+1} \tilde{\Phi}(e_n, \zeta_{n+1}) \quad (58)$$

with $\|e_N\| \leq \rho$ almost surely. Then $e_n \rightarrow 0$ almost surely,

$$\sum_{n=N}^\infty \gamma_{n+1} \|e_n\|^2 < \infty \quad \text{almost surely,} \quad (59)$$

and there are constants $c, C > 0$ such that

$$\mathbb{E} \|e_n\|^2 \leq C\gamma_n + Ce^{-c(t_n - t_N)}. \quad (60)$$

Proof. Let $P \succ 0$ solve $J^\top P + PJ = I$ and set $V(e) = e^\top Pe$. By decreasing ρ if necessary,

$$2e^\top P \tilde{\phi}(e) \geq c_1 \|e\|^2 \quad \text{for every } e. \quad (61)$$

Indeed, the linear part contributes $\|e\|^2$, the nonlinear part is $O(\|e\|^3)$ on $B_{2\rho}$, and it vanishes outside that ball. The cutoff noise vanishes outside $B_{2\rho}$ and has a bounded local second moment inside, so

$$\mathbb{E}[\|\tilde{\Phi}(e, \zeta)\|^2] \leq C(1 + \|e\|^2). \quad (62)$$

A one-step conditional expansion, with the $O(\gamma_{n+1}^2 \|e_n\|^2)$ term absorbed for all sufficiently large n , gives

$$\mathbb{E}[V(e_{n+1}) \mid \mathcal{F}_n] \leq (1 - c_2 \gamma_{n+1})V(e_n) + C\gamma_{n+1}^2. \quad (63)$$

Robbins–Siegmund’s theorem [4] implies convergence of $V(e_n)$ and (59). Since $\sum_n \gamma_n = \infty$, the only possible limit is zero.

Taking expectations in (63) gives $u_{n+1} \leq (1 - c_2 \gamma_{n+1})u_n + C\gamma_{n+1}^2$. Since $\gamma_n - \gamma_{n+1} = o(\gamma_n^2)$ for $\alpha < 1$, comparison with a sufficiently large multiple of γ_n , plus the homogeneous solution, yields $u_n \leq C\gamma_n + Ce^{-c(t_n - t_N)}$. Equivalence of V and $\|\cdot\|^2$ proves (60). $\square$

A.3 Raw-iterate CLT for the cutoff recursion

Let

$$\begin{aligned} \eta_{n+1}(e) &= \tilde{\Phi}(e, \zeta_{n+1}) - \tilde{\phi}(e), \\ \eta_{n+1}^0 &= \Phi(0, \zeta_{n+1}), \\ d_{n+1}(e) &= \eta_{n+1}(e) - \eta_{n+1}^0. \end{aligned}$$

The variables η_n^0 are i.i.d., centered, and have covariance Ω . The L^2 -continuity assumption implies

$$v(e) := \mathbb{E} \|d_1(e)\|^2 \longrightarrow 0 \quad \text{as } e \rightarrow 0. \quad (64)$$

The cutoff makes v bounded on $\mathbb{R}^q$.

Iteration of (58) gives

$$e_n = \Pi_{n,N} e_N - \sum_{k=N+1}^n \gamma_k \Pi_{n,k} \eta_k^0 - \sum_{k=N+1}^n \gamma_k \Pi_{n,k} d_k(e_{k-1}) - \sum_{k=N+1}^n \gamma_k \Pi_{n,k} \tilde{r}(e_{k-1}). \quad (65)$$

The first term is $o(\sqrt{\gamma_n})$ by (49). For the last term, Lemmas A.1 and A.3 give

$$\begin{aligned} \mathbb{E} \left\| \sum_{k=N+1}^n \gamma_k \Pi_{n,k} \tilde{r}(e_{k-1}) \right\|^2 &\leq C \sum_{k=N+1}^n \gamma_k e^{-c(t_n - t_k)} \mathbb{E} \|e_{k-1}\|^2 \\ &\leq C\gamma_n + Ce^{-c'(t_n - t_N)} = o(\sqrt{\gamma_n}). \end{aligned}$$

For the state-dependent martingale term, orthogonality gives

$$\frac{1}{\gamma_n} \mathbb{E} \left\| \sum_{k=N+1}^n \gamma_k \Pi_{n,k} d_k(e_{k-1}) \right\|^2 \leq \frac{C}{\gamma_n} \sum_{k=N+1}^n \gamma_k^2 e^{-c(t_n - t_k)} \mathbb{E} v(e_{k-1}).$$

By Lemma A.3, $e_k \rightarrow 0$ almost surely. Bounded convergence and (64) give $\mathbb{E} v(e_k) \rightarrow 0$. Splitting off finitely many early indices and applying (50) shows that the last display tends to zero. Hence

$$\gamma_n^{-1/2} e_n = - \sum_{k=N+1}^n \frac{\gamma_k}{\sqrt{\gamma_n}} \Pi_{n,k} \eta_k^0 + o_{\mathbb{P}}(1). \quad (66)$$

The covariance of the leading triangular array converges by Lemma A.2. By (51), its largest coefficient is $O(\sqrt{\gamma_n})$, while the sum of squared coefficient norms is bounded. For $B_{n,k} = \gamma_k \gamma_n^{-1/2} \Pi_{n,k}$, the $(2 + \delta)$ moment bound yields

$$\sum_k \mathbb{E} [\|B_{n,k} \eta_k^0\|^2 \mathbf{1}_{\{\|B_{n,k} \eta_k^0\| > \varepsilon\}}] \leq C_\varepsilon (\max_k \|B_{n,k}\|)^\delta \sum_k \|B_{n,k}\|^2 \longrightarrow 0.$$

The Lindeberg–Feller theorem and Cramér–Wold prove the raw CLT.

A.4 Averaging CLT for the cutoff recursion

From (58),

$$J e_{k-1} = -\frac{e_k - e_{k-1}}{\gamma_k} - \tilde{r}(e_{k-1}) - \eta_k^0 - d_k(e_{k-1}). \quad (67)$$

The nonlinear term is negligible because

$$\mathbb{E} \left[\frac{1}{\sqrt{n}} \sum_{k=N+1}^n \|\tilde{r}(e_{k-1})\| \right] \leq \frac{C}{\sqrt{n}} \sum_{k=N+1}^n \mathbb{E} \|e_{k-1}\|^2 = O(n^{1/2-\alpha}) = o(1). \quad (68)$$

The state-dependent martingale term is negligible in L^2 :

$$\mathbb{E} \left\| \frac{1}{\sqrt{n}} \sum_{k=N+1}^n d_k(e_{k-1}) \right\|^2 = \frac{1}{n} \sum_{k=N+1}^n \mathbb{E} v(e_{k-1}) \rightarrow 0 \quad (69)$$

by Cesàro's lemma.

Put $a_k = 1/\gamma_k$. Summation by parts gives

$$\sum_{k=N+1}^n a_k(e_k - e_{k-1}) = a_n e_n - a_{N+1} e_N - \sum_{k=N+1}^{n-1} (a_{k+1} - a_k) e_k. \quad (70)$$

Now $a_{k+1} - a_k = O(k^{\alpha-1})$ and $\mathbb{E} \|e_k\| = O(k^{-\alpha/2}) + O(e^{-ct_k})$. Consequently,

$$\frac{1}{\sqrt{n}} \sum_{k=N+1}^n \frac{e_k - e_{k-1}}{\gamma_k} = O_{L^1}(n^{(\alpha-1)/2}) + o(1) = o_{\mathbb{P}}(1). \quad (71)$$

Summing (67) therefore yields

$$\frac{1}{\sqrt{n}} \sum_{k=N}^{n-1} e_k = -J^{-1} \frac{1}{\sqrt{n}} \sum_{k=N+1}^n \eta_k^0 + o_{\mathbb{P}}(1). \quad (72)$$

The ordinary multivariate central limit theorem proves the averaging limit.

A.5 Transfer from the cutoff recursion to the original process

Return to Theorem 4.3. For each deterministic N , define

$$\pi_\rho(0) = 0, \quad \pi_\rho(e) = e \min\{1, \rho/\|e\|\} \quad (e \neq 0),$$

and start the cutoff recursion at

$$\widetilde{W}_N^{(N)} = w_0 + \pi_\rho(W_N - w_0),$$

using the same future innovations. Let

$$A_N = \{\text{the original recursion is exact for every } k \geq N\} \cap \left\{ \sup_{k \geq N} \|W_k - w_0\| \leq \rho \right\}.$$

The assumed eventual recursion and $W_n \rightarrow w_0$ imply $\mathbb{P}(A_N) \rightarrow 1$. On A_N , the clipped initial condition is W_N and the cutoff equals one along the future path, so induction gives $\widetilde{W}_k^{(N)} = W_k$ for every $k \geq N$.

For each fixed N , the cutoff recursion satisfies the raw and averaging CLTs above, with limits independent of N . The discrepancy probability is at most $\mathbb{P}(A_N^c)$ for the raw iterate. For the average, the finite prefix contributes $n^{-1/2} \sum_{k=0}^{N-1} (W_k - w_0) \rightarrow 0$ almost surely. The same coupling and (60) give

$$n^{-1/2} \sum_{k=0}^{n-1} \|W_k - w_0\|^2 \rightarrow 0 \quad \text{in probability,}$$

which is (37). Letting first $n \rightarrow \infty$ and then $N \rightarrow \infty$ transfers both CLTs. On the almost-sure union of the events A_N , the original tail equals a cutoff recursion, so (33) transfers as well. This proves Theorem 4.3.

References

- [1] M. Arnaiz Iglesias, A. R. Cetingoz, and N. Frikha, *Mirror Descent Algorithms for Risk Budgeting Portfolios*, arXiv:2411.12323, 2024.
- [2] V. Fabian, On asymptotic normality in stochastic approximation, *Annals of Mathematical Statistics* 39 (1968), 1327–1332.
- [3] B. T. Polyak and A. B. Juditsky, Acceleration of stochastic approximation by averaging, *SIAM Journal on Control and Optimization* 30 (1992), 838–855.
- [4] H. Robbins and D. Siegmund, A convergence theorem for nonnegative almost supermartingales and some applications, in *Optimizing Methods in Statistics*, Academic Press, 1971, 233–257.
- [5] H. J. Kushner and G. G. Yin, *Stochastic Approximation and Recursive Algorithms and Applications*, 2nd ed., Springer, 2003.